

the $\mathcal{H}_2$ norm of the error signal $\mathbf{e}$ is

$$\|\mathbf{e}\|_2 = \sqrt{\mathbf{e}^T \mathbf{e}}.$$

Let $\mathbf{e}_1$ and $\mathbf{e}_2$ be the error signals of the two parallel channels, then

$$\|\mathbf{e}\|_2 = \sqrt{\mathbf{e}_1^T \mathbf{e}_1 + \mathbf{e}_2^T \mathbf{e}_2}.$$

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